

Databricks + Fabric

A TANDEM DATA DELIVERY ARCHITECTURE

Combining data engineering power with unified SaaS business intelligence and democratization.

| What is Databricks?

Unified Lakehouse Architecture

Databricks is a unified, cloud-native PaaS that simplifies and democratizes data. It combines the performance of data warehouses with the flexibility of data lakes into a single "Lakehouse".

Spark Compute and Engineering

Powered by high-performance Spark engineering and Photon-accelerated compute.

Unity Catalog

Unity Catalog provides unified governance, security, and lineage across all data assets.

Workspace Environment

A collaborative environment for Data Engineering, SQL Analytics, and Machine Learning.



| What is Microsoft Fabric?

Unified SaaS Data Platform

Microsoft Fabric is an all-in-one analytics solution for enterprises that covers everything from data movement to data science, Real-Time Analytics, and business intelligence.

OneLake (The OneDrive for Data)

A single, unified, logical data lake for the entire organization, reducing data duplication and silos.

Experience-Based Analytics

Tailored experiences for every role: Data Factory for ETL, Synapse for Data Engineering, and Power BI for visualization.

Power BI

Power BI to visualize data sets directly from OneLake without moving or duplicating data.



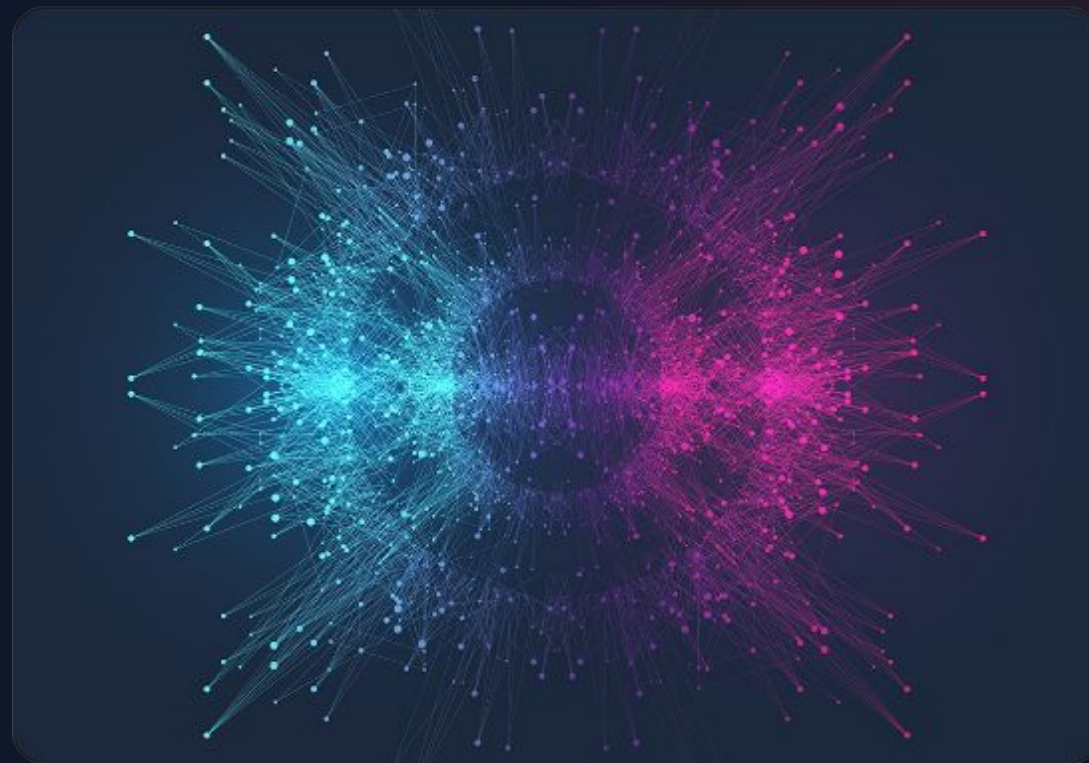
| The "Best of Breed" Synergy

Complementary Strengths

The tandem approach recognizes that **Engineers** and **Analysts** have different needs. This architecture solves for both without data duplication.

⚙️ **Databricks (ie; "engine")**: Best for high-performance Spark engineering, Photon-accelerated compute, and complex workflow lifecycles.

📦 **Fabric (ie; "storefront")**: Best for unified SaaS consumption, Direct Lake Power BI, and global data distribution across M365 (Dynamics, Power Platform, etc.)



Resolving Capability Ambiguity

Choosing the right tool for a specific **use**:

Capability	Databricks 	Fabric 
Notebooks	PySpark engineering, custom libraries, and git-integrated DevOps.	Collaborative data science and explorations for business analysts.
ETL/Jobs	Lakeflow Spark Declarative Pipelines, Databricks Jobs, and Photon-optimized workflows.	Visual, drag-and-drop orchestration via integrated Fabric Data Factory.
SQL Warehouse	Serverless SQL with TPC-DS scale performance for concurrent BI loads.	Fully managed SaaS SQL Analytics for quick-turn business queries.
Distribution	Operational SQL dashboards and custom data sharing (Delta/Open Sharing).	Enterprise-wide Power BI distribution and M365 app integration.

| Integration Key = Shortcuts!

Zero-Copy Virtualization

OneLake **Shortcuts** allow Fabric to "live mount" data stored in Databricks ADLS Gen2 paths without moving or copying the data.

Single Source of Truth: Point Fabric at the Gold tables maintained by Unity Catalog.

ACID Reliability: Fabric recognizes the Delta Log, ensuring consistent reads of Databricks-written data.

Cost Reduction: Eliminate duplicate storage costs and cross-tenant egress fees.

When to Shortcut?

- Data is already in ADLS Gen2/S3
- Direct connection wanted in Power BI
- You require multi-cloud flexibility (ADLS/S3 to OneLake)
- Storage-level security (RBAC) is established



Shortcut Example: 3-Step Technical Flow



1. Authenticate

Configure a Service Principal with rights to a data storage container.



2. Find Path

Locate the ADLS Gen 2 URL for the curated Delta tables managed by Unity Catalog.



3. Create Shortcut

In Fabric Lakehouse, set up the ADLS Gen 2 Shortcut for your chosen data.



Result: Databricks Delta tables appear instantly as Fabric-native objects, ready for SQL or Power BI Direct Lake mode.

Workload Orchestration: Fabric Pipelines & Databricks Workloads



BENEFITS OF DATA PIPELINE ORCHESTRATION

AUTOMATE
WORKFLOWS

ELIMINATE
SILOS

UNLOCK
DATA

TIME-TO-
INSIGHTS

Unified Orchestration Control

Data Factory in Fabric acts as the controller for cross-platform workflows.

The Databricks Activity: Drag-and-drop orchestration to trigger Notebooks, JARs, or DLT workloads.

Dynamic Parameters: Pass pipeline variables (e.g., date ranges) into Databricks widgets at runtime.

Managed Auth: Use Azure Entra ID for secure job invocation.

Governance: Databricks Unity Catalog

Local Engineering Control

Unity Catalog (UC) is the governance anchor for the Lakehouse, managing the "Engineering" perimeter.

Granular RBAC: Row and column-level security policies applied to Spark and SQL workloads.

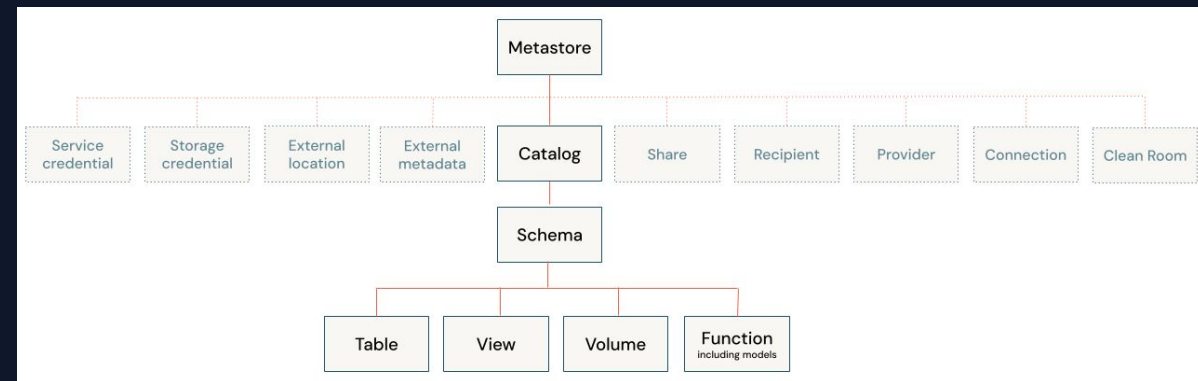
Lineage Tracking: Automated capture of data flows from raw files to curated tables.

External Locations: Governed access to the underlying storage containers in ADLS.



Why UC First?

By securing data in UC at the source, you ensure that any "Shortcut" into Fabric inherits a governed, cleansed dataset from the start.



| Enterprise Visibility: Microsoft Purview



The Global Data Map

Microsoft Purview acts as the **Enterprise Catalog**, bridging the gap between Databricks and the rest of the business. Security brought to 'edge device' of the user.

Unified Discovery: Allow business users to search for "product" datasets across the entire tenant.

Automated Classification: Detect PII and sensitive data patterns in Databricks tables.

Cross-Cloud Mapping: Map lineage from ADLS or S3 (Databricks) to Power BI (Fabric).

| Federated Governance



How **Unity Catalog (UC)** and **Purview** work in tandem:

The Scanner

The Purview Scanner connects to the UC Metastore APIs to ingest schemas, lineage, and tags automatically.

Policy Sync

Metadata tags applied in Purview can be used to inform masking policies within the Databricks SQL engine.

End-to-End

View a single graph showing a source file moving into DBx, into a Gold table, and out to a Fabric report.

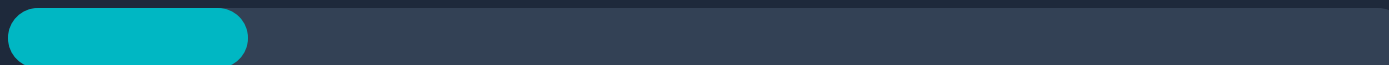
Power BI Performance Benchmarking



Modern report interactivity is defined by the underlying storage mode. Direct Lake has made improvements almost matching Import performance for most enterprise scales.

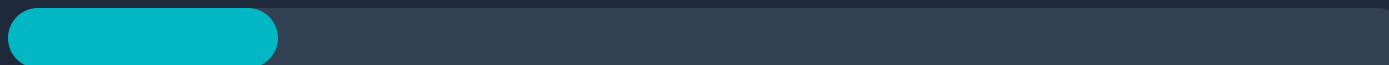
Import Mode

Data is loaded into VertiPaq memory for peak performance.



Direct Lake

Zero-copy memory mapping directly to OneLake Delta files.



Direct Query

Latency is 100% dependent on the backend SQL engine call.



Comparison based on sub-second report interactivity requirements for modern executive dashboards.

| Medallion...?



Medallion Layer	Platform	Reasoning
Bronze (Raw)	Databricks	High-volume file ingestion. Schema inference. Declarative pipelines, Auto Loader, Unity Catalog governance.
Silver (Cleansed)	Databricks	Photon-accelerated Spark for deduplication, PII masking, and validation.
Gold (Curated)	Microsoft Fabric	Architectural hand-off: Silver tables shared into Gold - Gold tables shared via OneLake.
Distribution (BI)	Microsoft Fabric	Semantic modeling and Direct Lake BI for the global executive audience.

Strategic Approach



Orchestrate



Foundation

Engineering, Delta Tables



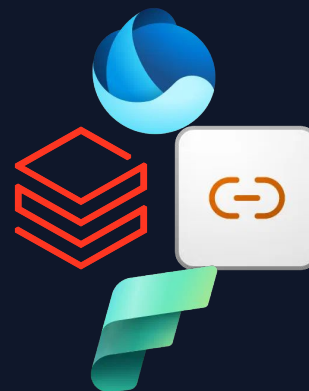
Govern

Unity Catalog & Purview



Integrate

OneLake Shortcuts

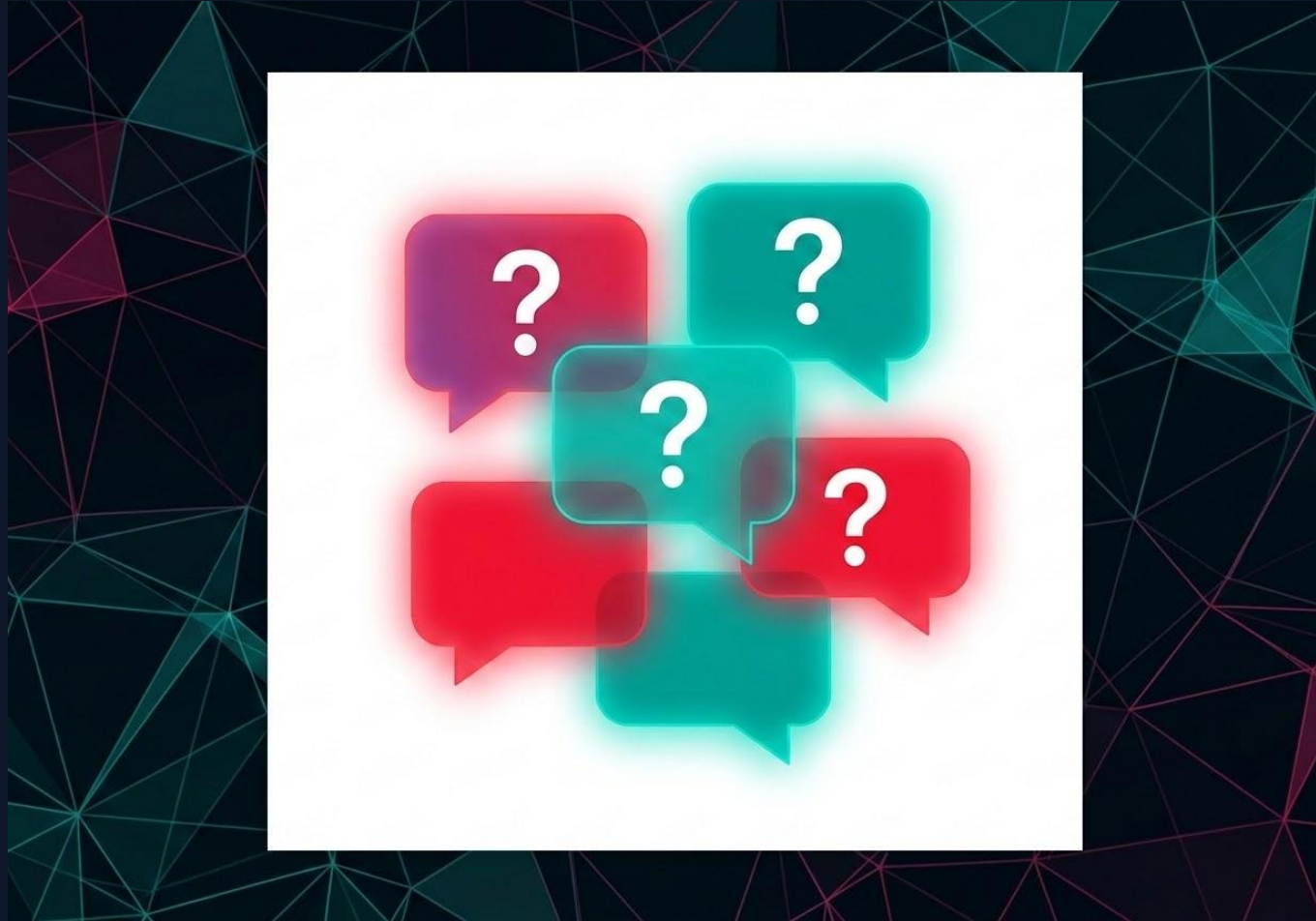


Serve

Power BI



| Questions, Discussion



(Yes, this image is gen AI slop)